

Zhaokai Wang (王肇凯)

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HIGHLIGHTS

- **Core Contributor of InternVL Series:** Key member of open-source models InternVL3.5, InternVL-U and Mono-InternVL at Shanghai AI Laboratory.
- **Prolific Publications:** Published over 15 papers in top-tier conferences/journals including TPAMI, NeurIPS, ICML, CVPR, ICCV, with 10 as first or co-first author.
- **Research Impact:** Accumulated 2500+ citations. Receive Best Paper Runner-up at NeurIPS 2025, Best Paper Award at ACM Multimedia 2021, and three workshop best paper awards.

EDUCATION

Shanghai Jiao Tong University

Ph.D. Student (Joint Program with Shanghai AI Laboratory)

09/2022 - 06/2027 (Expected)

Advisor: Prof. Jifeng Dai

Peking University

Double Bachelor Degree in Economics

09/2019 - 06/2022

Beihang University

B.Eng in Computer Science

09/2018 - 06/2022

Overall GPA: 3.80

EXPERIENCE

OpenGVLab (InternVL Team), Shanghai AI Laboratory

Research Intern, collaborated with Prof. Jifeng Dai and Dr. Xizhou Zhu

Shanghai

11/2022 - 11/2025

Fundamental Vision Group, SenseTime

Research Intern, collaborated with Prof. Jifeng Dai and Dr. Xizhou Zhu

Beijing

02/2022 - 10/2022

Sea AI Lab

Research Intern, collaborated with Prof. Shuicheng Yan and Dr. Jibin Wu

Remotely at Beijing

08/2021 - 02/2022

CoLab, Institute of Artificial Intelligence at Beihang University

Research Intern, collaborated with Prof. Si Liu

Beijing

08/2019 - 06/2022

AWARDS

- Main Conference
 - Best Paper Runner Up Award**, NeurIPS 2025 2025
 - Best Paper Award**, ACM Multimedia 2021 2021
- Workshops and Competitions
 - Best Paper Award**, ICML AI4MATH Workshop 2025 2025
 - Best Paper Award**, IJCAI MKLM Workshop 2025 2025
 - Best Paper Award**, KDD Urban Computing Workshop (UrbComp) 2024 2024
 - Best Video Award, IJCAI 2021 Video Competition 2021
 - First Place, CVPR 2020 TextCaps Challenge 2020

PUBLICATIONS

(*: Co-first authors.)

1. GenExam: A Multidisciplinary Text-to-Image Exam
Zhaokai Wang, Penghao Yin, Xiangyu Zhao, Changyao Tian, Yu Qiao, Wenhai Wang, Jifeng Dai, Gen Luo
ICML 2026
2. Parameter-Inverted Image Pyramid Networks for Visual Perception and Multimodal Understanding
Zhaokai Wang, Xizhou Zhu, Xue Yang, Gen Luo, Hao Li, Changyao Tian, Wenhan Dou, Junqi Ge, Lewei Lu, Yu Qiao, Jifeng Dai
TPAMI 2025
3. DisciplineGen-1M: A Large-Scale Dataset for Multidisciplinary Visual Generation and Editing
Zhaokai Wang, Mingxin Liu, Zirun Zhu, Ziqian Fan, Yiguo He, Mohan Zhang, Leyao Gu, Xiangyu Zhao, Ning Liao, Shaofeng Zhang, Xuanhe Zhou, Zhihang Zhong, Junchi Yan, Xue Yang
Preprint

4. Vision-to-Music Generation: A Survey
Zhaokai Wang, Chenxi Bao, Le Zhuo, Jingrui Han, Yang Yue, Yihong Tang, Victor Shea-Jay Huang, Yue Liao
ISMIR 2025
5. Confidence-aware Non-repetitive Multimodal Transformers for TextCaps
Zhaokai Wang, Renda Bao, Qi Wu, Si Liu
AAAI 2021
6. InternVL3.5: Advancing Open-Source Multimodal Models in Versatility, Reasoning, and Efficiency
Weiyun Wang*, Zhangwei Gao*, Lixin Gu*, Hengjun Pu*, Long Cui*, Xingguang Wei*, Zhaoyang Liu*, Linglin Jing*, Shenglong Ye*, Jie Shao*, Zhaokai Wang*, Zhe Chen*, Hongjie Zhang, Ganlin Yang, Haomin Wang, Qi Wei, Jinhui Yin, Wenhao Li, Erfei Cui, Guanzhou Chen, Zichen Ding, Changyao Tian, Zhenyu Wu, Jingjing Xie, Zehao Li, Bowen Yang, Yuchen Duan, Xuehui Wang, Zhi Hou, Haoran Hao, Tianyi Zhang, Songze Li, Xiangyu Zhao, Haodong Duan, Nianchen Deng, Bin Fu, Yinan He, Yi Wang, Conghui He, Botian Shi, Junjun He, Yingdong Xiong, Han Lv, Lijun Wu, Wenqi Shao, Kaipeng Zhang, Huipeng Deng, Biqing Qi, Jiaye Ge, Qipeng Guo, Wenwei Zhang, Songyang Zhang, Maosong Cao, Junyao Lin, Kexian Tang, Jianfei Gao, Haiyan Huang, Yuzhe Gu, Chengqi Lyu, Huanze Tang, Rui Wang, Haijun Lv, Wanli Ouyang, Limin Wang, Min Dou, Xizhou Zhu, Tong Lu, Dahua Lin, Jifeng Dai, Weijie Su, Bowen Zhou, Kai Chen, Yu Qiao, Wenhao Wang, Gen Luo
Technical Report
7. InternVL-U: Democratizing Unified Multimodal Models for Understanding, Reasoning, Generation and Editing
Changyao Tian*, Danni Yang*, Guanzhou Chen*, Erfei Cui*, Zhaokai Wang*, Yuchen Duan*, Penghao Yin*, Sitao Chen, Ganlin Yang, Mingxin Liu, Zirun Zhu, Ziqian Fan, Leyao Gu, Haomin Wang, Qi Wei, Jinhui Yin, Xue Yang, Zhihang Zhong, Qi Qin, Yi Xin, Bin Fu, Yihao Liu, Jiaye Ge, Qipeng Guo, Gen Luo, Hongsheng Li, Yu Qiao, Kai Chen, Hongjie Zhang
Technical Report
8. GRADE: Benchmarking Discipline-Informed Reasoning in Image Editing
Mingxin Liu*, Ziqian Fan*, Zhaokai Wang* (*Project Lead*), Leyao Gu*, Zirun Zhu*, Yiguo He, Yuchen Yang, Changyao Tian, Xiangyu Zhao, Ning Liao, Shaofeng Zhang, Qibing Ren, Zhihang Zhong, Xuanhe Zhou, Junchi Yan, Xue Yang
ECCV 2026
9. Mono-InternVL-1.5: Towards Cheaper and Faster Monolithic Multimodal Large Language Models
Gen Luo, Wenhan Dou, Wenhao Li, Zhaokai Wang, Xue Yang, Changyao Tian, Hao Li, Weiyun Wang, Wenhao Wang, Xizhou Zhu, Yu Qiao, Jifeng Dai
Technical Report
10. Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?
Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, Gao Huang
NeurIPS 2025 Best Paper Runner Up Award (Top 1 score among 21575 submissions) & ICML 2025 AI4MATH Workshop Best Paper Award (2/172)
11. Video Background Music Generation with Controllable Music Transformer
Shangzhe Di, Zeren Jiang, Si Liu, Zhaokai Wang, Leyan Zhu, Zexin He, Hongming Liu, Shuicheng Yan
ACM Multimedia 2021 Best Paper Award (1/1942)
12. Sparkle: Mastering Basic Spatial Capabilities in Vision Language Models Elicits Generalization to Composite Spatial Reasoning
Yihong Tang*, Ao Qu*, Zhaokai Wang*, Dingyi Zhuang*, Zhaofeng Wu, Wei Ma, Shenhao Wang, Yunhan Zheng, Zhan Zhao, Jinhua Zhao
EMNLP 2025 Findings & IJCAI 2025 MKLM Workshop Best Paper Award
13. Mono-InternVL: Pushing the Boundaries of Monolithic Multimodal Large Language Models with Endogenous Visual Pre-training
Gen Luo*, Xue Yang*, Wenhan Dou*, Zhaokai Wang*, Jiawen Liu, Jifeng Dai, Yu Qiao, Xizhou Zhu
CVPR 2025
14. SynerGen-VL: Towards Synergistic Image Understanding and Generation with Vision Experts and Token Folding
Hao Li, Changyao Tian, Jie Shao, Xizhou Zhu, Zhaokai Wang, Jinguo Zhu, Wenhan Dou, Xiaogang Wang, Hongsheng Li, Lewei Lu, Jifeng Dai
CVPR 2025
15. Parameter-Inverted Image Pyramid Networks
Xizhou Zhu*, Xue Yang*, Zhaokai Wang*, Hao Li, Wenhan Dou, Junqi Ge, Lewei Lu, Yu Qiao, Jifeng Dai
NeurIPS 2024 Spotlight (Top 10 scores among 15671 submissions, Top 2 in Computer Vision)
16. ITINERA: Integrating Spatial Optimization with Large Language Models for Open-domain Urban Itinerary Planning
Yihong Tang*, Zhaokai Wang*, Ao Qu*, Yihao Yan*, Zhaofeng Wu, Dingyi Zhuang, Jushi Kai, Kebing Hou, Xiaotong Guo, Jinhua Zhao, Zhan Zhao, Wei Ma
EMNLP 2024 Industry Track & KDD 2024 UrbComp Workshop Best Paper Award

17. Auto MC-Reward: Automated Dense Reward Design with Large Language Models for Minecraft
Hao Li*, Xue Yang*, Zhaokai Wang*, Xizhou Zhu, Jie Zhou, Yu Qiao, Xiaogang Wang, Hongsheng Li, Lewei Lu, Jifeng Dai
CVPR 2024
18. Video Background Music Generation: Dataset, Method and Evaluation
Le Zhuo*, Zhaokai Wang*, Baisen Wang*, Yue Liao, Chenxi Bao, Stanley Peng, Songhao Han, Aixi Zhang, Fei Fang, Si Liu
ICCV 2023

ADDITIONAL INFORMATION

Talks

- 2026.1: *From Frames to Beats: Progress and Challenges in Video-to-Music Generation*. At Centre for Digital Music (C4DM) of Queen Mary University of London.
- 2025.4: *Mono-InternVL: Pushing the Boundaries of Monolithic Multimodal Large Language Models with Endogenous Visual Pre-training*. At Open Multimodal Gathering Workshop hosted by NUS ShowLab.

Activities

- **Conference Reviewer:** ICCV 2023 & 2025, ECCV 2024 & 2026, CVPR 2024 & 2025, EMNLP 2024, NeurIPS 2024 & 2025, ICLR 2025, ICML 2025, AAAI 2025.
- **Journal Reviewer:** TPAMI, ACM Computing Survey.
- **Teaching Assistant:** Fundamentals of Computers (2021), Software Engineering (2022).

Skills

- **Programming languages:** Python, C/C++, Java, JavaScript.
- **Scientific packages:** Numpy, Pytorch, Tensorflow.
- **English:** TOEFL 111 (S26) , GRE 327+4.0, CET-4 669, CET-6 612.